Loan Default Prediction

Comparing Classic ML and Neural Networks under Class Imbalance

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Home Credit Default Risk · Kaggle

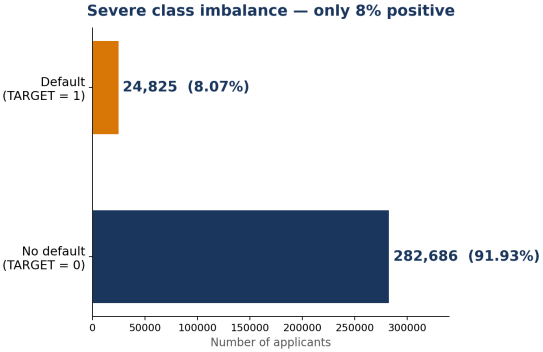
Problem & Motivation

Task	**Binary classification** — predict whether a Home Credit applicant defaults (`TARGET` $\in \{0, 1\}$).
Data	307,511 applicants $\times$ 122 features (Kaggle, real-world).
Challenge	Severe class imbalance — only **8.07%** default; a "predict-no-default" baseline scores 92% accuracy with 0% recall.
Approach	Compare classic ML (LR, RF, HGBC) and neural networks (Shallow / Deep MLP) under the same pipeline; then study imbalance-handling on the best NN.
Metric	ROC AUC (accuracy is misleading on imbalanced data).

Research question: do classic shallow learners and neural networks

Dataset · Home Credit Default Risk

Structure	Tabular (row = one applicant)
Observations	307,511 train / 48,744 te
Features	122 columns (102 numeric, 16 categorical)
Target	TARGET $\in \{0, 1\}$ 1 = default, 0 = repaid
Imbalance	default rate = **8.07%**



Feature themes

- Demographics, employment, credit-bureau summaries, requested loan amount

Classic ML Models

Model (family)	Best hyperparameters	Val AUC
Logistic Regression *(linear)*	`C`=0.1; `class_weight=balanced`	0.7487
Random Forest *(bagging)*	`n_estimators`=200; `min_samples_split`=100; `class_weight=balanced`	0.7470
HistGradientBoosting *(boosting)*	`learning_rate`=0.05; `max_iter`=200; `min_samples_leaf`=100	**0.7595**

Key takeaways

- `class_weight='balanced'` on LR / RF — handles imbalance directly in the loss.
- HGBC wins by ∼1 AUC point — boosted trees usually dominate tabular data.
- RF overfits at `leaf=1`: train AUC 0.96 vs val 0.747; regularized leaf wins.

Neural Network Models

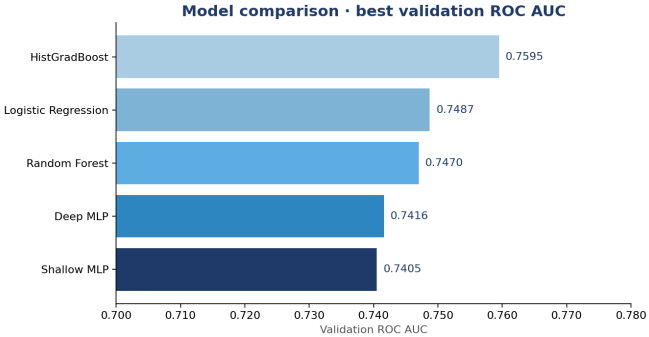
Architecture	Hidden layers	Parameters	Val AUC
Shallow MLP	$(64,)$	$\sim$15.7K	0.7405
Deep MLP	$(128, 64, 32)$	$\sim$42K	**0.7416**

Both: ReLU · Adam · early stopping. Tuned over `alpha` $\in \{10^{-4}, 10^{-3}\}$, `lr` $\in \{0.001, 0.005\}$. Best for both: $\alpha = 10^{-3}$, `lr` $= 0.005$.

Key takeaways

- Depth doesn't help: 0.7405 vs 0.7416 (within noise).
- Tripling parameters bought nothing.
- Both NNs land below the classic learners.

Combined Leaderboard



Model	Val AUC
HistGradBoost	**0.7595**
Logistic Reg.	0.7487
Random Forest	0.7470
Deep MLP	0.7416
Shallow MLP	0.7405

Tree boosting wins by $\sim$1 AUC point.

Imbalance Handling on the Best MLP

Method	Threshold	ROC AUC	PR AUC	Precision	Recall	F1
Baseline	0.500	0.745	0.222	0.00	0.00	0.00
SMOTE	0.500	**0.648**	0.134	0.16	0.21	0.18
Threshold tuning ⋆	0.164	0.745	0.222	0.24	**0.40**	**0.30**

Key takeaways

- Baseline at 0.5 cutoff: AUC 0.745 but recall **0%** (no prediction crosses 0.5).
- SMOTE hurts: AUC drops 0.10.
- Threshold tuning at 0.164 lifts recall to **40%**.

Choosing the threshold matters more than rebalancing the data.

Conclusions

What we learned

- HistGradientBoosting wins overall (val AUC = **0.7595**).
- Neural networks tie LR / RF — depth alone does not help on tabular data.
- Threshold tuning lifts MLP recall from **0% to 40%**; SMOTE hurt AUC.

Limitations

- Only main `application_*.csv` tables (auxiliary tables typically add several AUC points).
- `PredefinedSplit` is fast but optimistic; full k-fold CV would tighten CIs.

Recording

<<INSERT RECORDING LINK BEFORE SUBMISSION>>

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